

From schematic calibration to real layout: know when a decision is trustworthy

## 1 | A reusable experiment

A small calibrated offset is not enough. A comparator can decide the wrong way, miss its deadline, or change behavior after parasitic extraction.

This notebook makes the complete decision map, costs and failure boundaries reproducible instead of showing only one favorable nominal waveform.

### Circuit and design choices

Established StrongARM topology; 27 transistor instances.  
LVT main/auxiliary input path; standard-VT latch and switches.  
Physical source-gated trim: sign plus four magnitude bits.  
Nine declared candidates, followed by an explicit lower-energy control. No universal best-design claim.

### Review it in three steps

1. Inspect the source-matched circuit guide.
2. Change the input band, deadline and calibration policy.
3. Compare actual GDS, DRC/LVS controls, extracted RC and saved waveforms.

[Open the current submission in Colab](#)

Comparator\_Atlas.ipynb

Run all: checked evidence, fresh tables and waveform measurements. Live SPICE and the full campaign are explicit optional modes.  
Source, license, references and CI are in official PR #195.

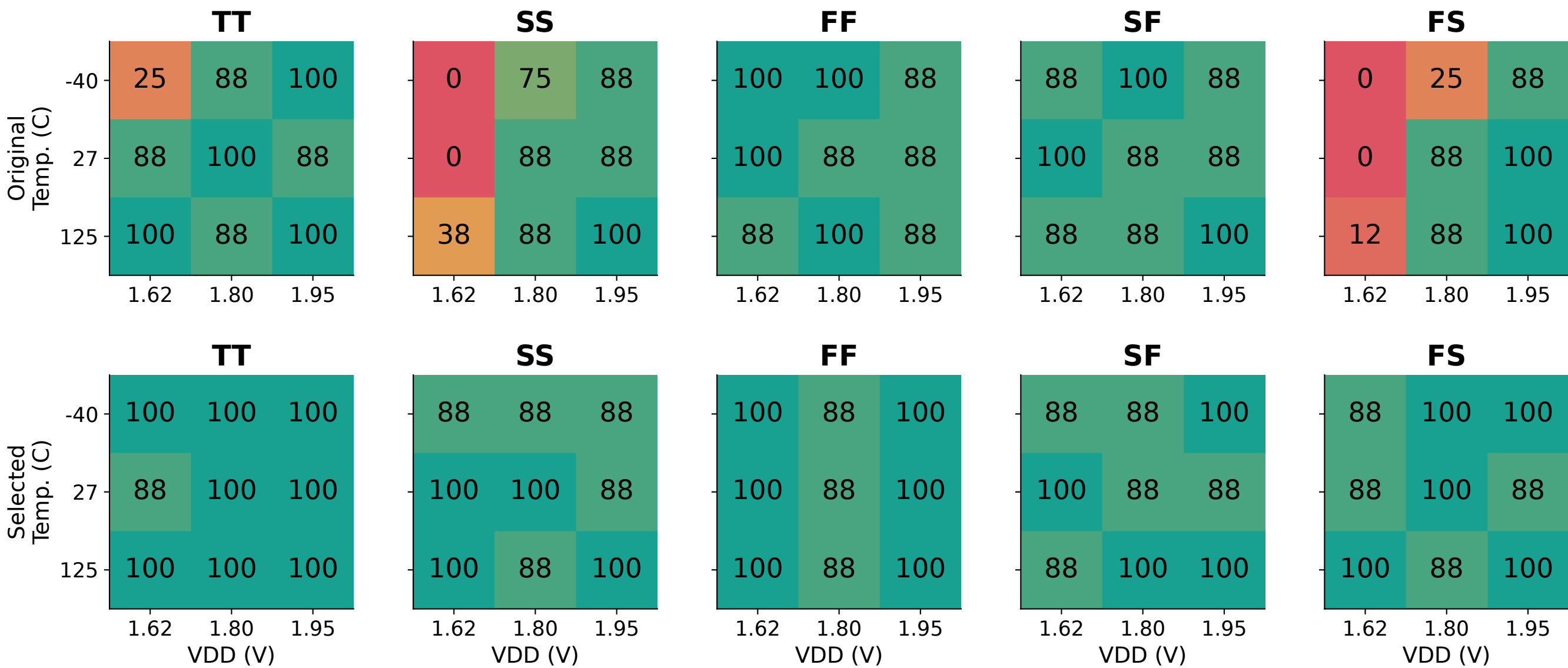
Finite sampled evidence, not silicon or foundry signoff. Core energy excludes external drivers and calibration infrastructure.

Established circuitry: Razavi, 2015, DOI 10.1109/MSSC.2015.2418155; Li, Xu & Iizuka, 2022, DOI 10.1007/s10470-022-01992-6.

Original code: MIT. SKY130 keeps Apache-2.0. GitHub Copilot assistance disclosed. Source, models and reproduction instructions are linked in the notebook.

## 2 | Schematic coverage is not free

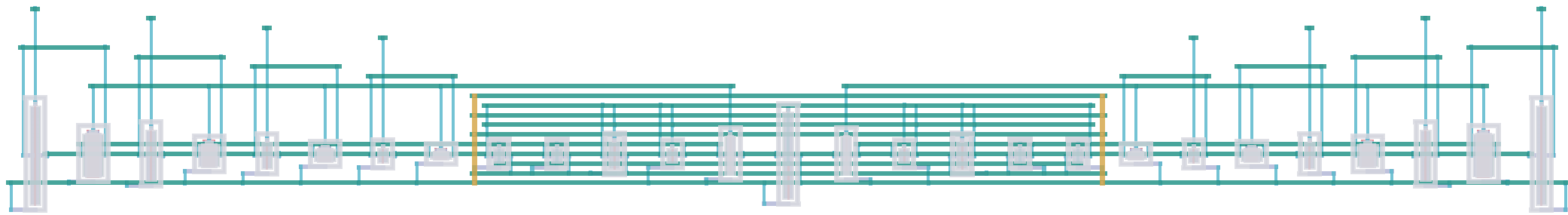
Same local calibration; 1 ns; sampled |input| >= 1 mV; +4% branch-width stress



**All 49 conditions: 79.1% -> 94.9% correct | 143.0 -> 251.5 fJ/cycle**

Above: 45 PVT combinations; aggregate also includes four nominal stress controls. Percentages are grid coverage, not yield.

## 3 | Real layout, extracted RC, explicit limits



**Actual 27-device GDS | DRC 0 + LVS / negative controls | 2207.088 um2**

**Nominal layout, code zero: RC 12/20 at 1 ns | 20/20 at retained 2 ns**

Five conditions x four signed inputs, not full 45-PVT or continuous-range qualification. The original 1 ns pilot fails; 2 ns is explicitly post-hoc characterization.

**Matched TT +/-3 mV, prior legal balanced -> repaired compact RC:**

**0.843 -> 0.645 ns; 520.8 -> 425.4 fJ (schematic: 244.1 fJ)**